

INHA UNIVERSITY TASHKENT
DEPARTMENT OF CSE & ICE
FALL SEMESTER 2024
SOC 3010 - OPERATING SYSTEM
TERM PROJECT PROPOSAL
DOCTORS RESERVATION SYSTEM
FOR PATIENTS

Submitted by

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SOC3010 OS TERM PROJECT FALL 2023

TERM PROJECT PROPOSAL PREPARATION GUIDELINES

1. **TERM PROJECT MUST BE IMPLEMENTED ON LINUX SYSTEM AND NOT ON ANY OTHER SYSTEM**
2. **GRADE '0' WILL BE AWARDED IF IT IS IMPLEMENTED ON ANY OTHER SYSTEM (LIKE WINDOWS OR ON OTHER OS SYSTEM)**
3. **OS TERM PROJECT PROPOSAL MUST BE PREPARED USING THE TERM PROJECT PROPOSAL TEMPLATE GIVEN AT THE ECLASS PORTAL**
4. **ONLY THE TERM PROJECT PROPOSAL PREPARED USING THE TEMPLATE GIVEN AT THE ECLASS WILL BE ACCEPTED.**
5. **TERM PROJECT PROPOSAL PREPARED USING ANY OTHER FORMAT WILL BE REJECTED AND GIVEN A GRADE 0.**
6. **A HARDCOPY OF THE PROPOSAL DOCUMENT MUST BE HANDED IN BY THE TEAM LEADER BY VISITING MY OFFICE ON MONDAY 21ST OCTOBER AFTERNOON ANYTIME BETWEEN 2PM & 5PM.**
7. **EVERY MEMBER OF THE TEAM INCLUDING THE TEAM LEADER MUST UPLOAD THE SOFTCOPY OF THE TERM PROJECT PROPOSAL DOCUMENT AT THE ECLASS PORTAL BY MONDAY 21ST OCTOBER 2024 WITHOUT FAIL.**
8. **LAST DATE FOR SUBMISSION OF THE TERM PROJECT PROPOSAL DOCUMENT AT THE ECLASS PORTAL BY EVERY MEMBER IS MONDAY, 21ST OCTOBER 2024.**
9. **NO SUBMISSION AT ECLASS PORTAL WILL LEAD TO GRADE 0 FOR THE TERM PROJECT.**
10. **PLEASE NOTE THAT WITHOUT THE TERM PROJECT PROPOSAL, TEAMS WILL NOT BE ALLOWED TO CARRY OUT THEIR TERM PROJECTS AND WILL BE GIVEN A GRADE 0 FOR THE TERM PROJECT.**
11. **LATE SUBMISSIONS ARE NOT ENTERTAINED, ADHERE TO THE DEADLINE STRICTLY**
12. **ALL INSTRUCTIONS REGARDING PREPARATION OF THE TERM PROJECT PROPOSAL SECTIONS ARE GIVEN IN THE PROPOSAL TEMPLATE SECTIONS ITSELF IN RED COLOR TEXT.**
13. **ANY STATEMENT IN RED IN THE TERM PROJECT PROPOSAL DOCUMENT INDICATES THE INSTRUCTIONS FOR YOU TO WRITE THE PROPOSAL. BUT THESE INSTRUCTIONS WRITTEN IN RED SHOULD BE REMOVED FROM THE FINAL PROPOSAL DOCUMENT WHICH YOU SUBMIT.**
14. **DO NOT WRITE YOUR PROPOSAL IN RED**
15. **THE BRIEF DESCRIPTION OF THE REPORT LAYOUT IS GIVEN BELOW:**
SECTION HEADING FONT – Times New Roman, 14pt
SECTION TEXT FONT – Times New Roman, 12pt
ALL FONTS AND SIZES ARE CLEARLY SET IN THE PROPOSAL DOCUMENT AND NOT TO BE MODIFIED
YOU ARE REQUIRED TO JUST ENTER THE TEXT IN ALL THE SECTIONS, INCLUDE REQUIRED FIGURES, CHARTS, IMAGES, TABLES AND SCREEN SHOTS.
IN THE FACING SHEET OF THE PROPOSAL YOU SHOULD WRITE THE PROJECT TITLE IN BOLD & CAPITAL, PROJECT LEADER & MEMBER NAMES WITH IDs & SECTION Nos. WITH PROPER ALIGNMENT.
YOU ARE REQUIRED TO GO THROUGH THE TERM PROJECT PROPOSAL TEMPLATE GIVEN AT THE ECLASS PORTAL CORRECTLY AND CAREFULLY
THE TERM PROJECT PROPOSAL DOCUMENT CONTAINS THE FOLLOWING SECTIONS AND INSTRUCTIONS ARE GIVEN IN RED IN EACH SECTION BELOW THE SECTION HEADING WHAT TO INCLUDE IN EACH SECTION.

BUT THESE INSTRUCTIONS IN RED SHOULD BE REMOVED FROM THE FINAL DOCUMENT BEFORE UPLOADING OR BEFORE TAKING PRINTOUT FOR SUBMISSION.

DO NOT WRITE YOUR REPORT IN RED.

TITLE OF THE PROJECT:

DOCTORS RESERVATION SYSTEM FOR PATIENTS

PROBLEM STATEMENT:

In today's healthcare system, patients face numerous challenges when seeking medical care. The manual appointment booking systems are prone to errors, double bookings, and lack real-time availability information, leading to inefficient use of medical resources and frustrating experiences for patients. Once at the hospital, patients often endure long queues and extended waiting times, which can be particularly distressing when dealing with health concerns. This time wastage is compounded when patients discover that their needed specialist is unavailable at that facility, forcing them to travel to another hospital and restart the entire waiting process. The current system not only strains hospital resources but also places an unnecessary burden on patients who may already be dealing with health issues. There is a clear need for a centralized, automated system that can efficiently manage doctor appointments while providing real-time information about specialist availability across different facilities, ensuring a seamless experience for both patients and healthcare providers.

ABSTRACT :

This project implements a Doctors Reservation System for Patients (DRSP) as a distributed client-server application for Linux platforms. The system provides efficient medical appointment scheduling and management through a three-layer architecture: a GTK+3-based graphical user interface, socket-based network communication layer, and MySQL database backend. The application serves three distinct client types: patients, doctors, and system administrators.

Patients can register with their personal information (including name, age, address, phone number, citizenship, and optional email), select their desired medical specialty, and book appointments with available doctors across different hospitals. The system provides real-time availability checking and appointment scheduling functionality. Doctors can manage their schedules, mark their unavailable time slots, and access patient histories. Additionally, the system implements a dynamic waitlist feature that automatically manages appointment cancellations and patient priority queuing.

The system administrator has comprehensive access to monitor system operations, manage database records, and oversee user activities through a dedicated connection monitoring window. The implementation utilizes C programming for inter-process communication, ensuring data exchange between distributed clients and the centralized server. All client operations, including registration, authentication, and appointment management, are processed through secure socket connections, with data persistently stored in the server-side database.

This project demonstrates practical implementation of operating system concepts including process management, inter-process communication, and network programming, while providing a valuable solution for doctor-patient appointment management.

INTRODUCTION

Healthcare appointment management is a critical aspect of medical service delivery. The traditional methods of appointment booking through phone calls or in-person visits are time-consuming and often result in scheduling conflicts. Our Doctors Reservation System aims to modernize this process by providing a digital platform that connects patients with healthcare providers efficiently.

The system addresses several key challenges:

- Streamlining the appointment booking process
- Reducing waiting times and scheduling conflicts
- Providing real-time availability information
- Managing patient records and appointment histories
- Enabling efficient communication between patients and doctors

Our solution implements a three-tier architecture:

1. GUI front-end
2. Middle layer handling communication between client and server, and between front-end and back-end
3. Database backend for data records

OVERALL PROJECT OVERVIEW :

System Components:

1. Client-Side Components:

a) Patient Interface:

- Registration and login system
- Information about Hospital, Specialities, and Doctors
- Hospital, Specialities, and Doctors selection
- Appointment booking and cancelling interface
- Booking into waitlist
- Case history information
- Schedule information of any doctor in daily and hourly time slots
- Notifications about appointment cancellation, approval or waitlist update
- Prescription provided by the Doctor

b) Doctor Interface:

- Appointment and waitlist management
- Case history of their patients
- Waitlist management
- Time slot allocation for various purposes

c) Admin Interface:

- System monitoring
- User management
- Full database access
- Real time connection window full access
- Password hardcoded into server code

2. Server-Side Components:

a) Connection Handler:

- Multi-client connection acceptance and management

- Concurrent user handling
 - Request routing system
 - Active connection maintenance
- b) Request Processor:
- Client request handling (appointments, cancellations)
 - Input validation system
 - Database communication interface
 - Client response management
- c) Authentication Module:
- User credential verification
 - Session management system
 - Login/logout processing
 - Data encryption handling
- d) Real-time Monitor:
- Connection tracking system
 - Server status monitoring
 - Activity logging mechanism
 - Error detection and reporting
3. Database Components:
- Different clients' profile information (including login and password)
 - Case history records
 - Appointment records
 - Doctors' schedule records
 - Information about hospitals
 - Information about doctors
 - Information about specialities

- Waitlist information
4. Security:
- Encryption of passwords

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PROJECT DESIGN AND IMPLEMENTATION:

1 Frontend Development (GUI):

- Implementation using GTK+3 with Glade
- Responsive user interfaces for all three user types
- Form validation and error handling
- Real-time updates for appointment availability

2 Middle Layer:

- Socket programming in C for client-server communication
- Multi-threaded server to handle multiple concurrent connections
- Implementation of business logic and data validation
- Secure data transmission protocols

3 Backend Development:

- MySQL database implementation
- Stored procedures for common operations
- Data integrity and consistency maintenance
- Backup and recovery mechanisms

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CONCLUSION:

The Doctors Reservation System represents a significant advancement in healthcare appointment management, addressing critical challenges in the current medical scheduling landscape. Through its three-tier architecture and comprehensive feature set, the system offers several key benefits:

1. Enhanced Efficiency:

- Streamlined appointment booking process reduces administrative overhead
- Real-time availability updates minimize scheduling conflicts
- Automated waitlist management optimizes resource utilization

2. Improved User Experience:

- Intuitive GTK+3-based interface for all user types
- Reduced waiting times for patients
- Simplified schedule management for healthcare providers
- Comprehensive administrative oversight capabilities

3. Technical Innovation:

- Robust client-server architecture using socket programming
- Secure data handling and transmission
- Scalable database design with MySQL
- Multi-threaded processing for concurrent user handling

4. Healthcare Impact:

- Better resource allocation across medical facilities
- Enhanced communication between patients and healthcare providers
- Improved access to specialist care through centralized booking
- Digital record keeping for better continuity of care

The system's modular design demonstrates our understanding of key operating system concepts including process management, inter-process communication, and network programming. Through practical implementation of these concepts, we've created a functional system that showcases both technical proficiency and real-world applicability. The successful completion of this project will provide valuable hands-on experience in developing multi-threaded, distributed applications on Linux systems.

PROJECT ACTION PLAN :

Term Project Tasks Breakup Weekwise

Sl. No	Duration	Specify Tasks to be carried out	Team Members assigned to do this task
1.	9 th Week 4 th Nov, to 10 th Nov. 2024	Frontend Team: Initial GUI design, design creation, GTK+3 setup Middle Layer Team: Socket programming research, Basic client-server architecture design	Frontend: Sabitov Samandar, Kobiljonov Davlatbek

		Backend Team: Database architecture designing	Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon
2.	10th Week 11th Nov. to 17th Nov. 2024	Frontend Team: Patient interface development, testing it with the database. Middle Layer Team: Basic socket connection implementation, Multi-client handling setup. Backend Team: Database creation, Initial table implementation, Basic queries development	Frontend: Sabitov Samandar, Kobiljonov Davlatbek Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon
3.	11th Week 18th Nov. to 24th Nov. 2024	Frontend Team: Doctor's interface development. Testing it with the database. Middle Layer Team: Authentication system, Session management, password encryption implementation. Backend Team: Stored procedures, Trigger implementation, Data validation rules	Frontend: Sabitov Samandar, Kobiljonov Davlatbek Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon
4.	12th Week 25th Nov. to 1st Dec. 2024	Frontend Team: System administrator's interface development. Testing it with the database. Middle Layer Team: Data transmission protocols, Connection monitoring system. Backend Team: Query optimization, Index creation, Backend API development	Frontend: Sabitov Samandar, Kobiljonov Davlatbek Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon

5.	13th Week 2nd Dec. to 8th Dec. 2024	Frontend Team: Integration of front-end layers. Middle Layer Team: Integration of middle layer with backend, testing for edge cases. Backend team: Integration of backend with middlelayer, testing for edge cases.	Frontend: Sabitov Samandar, Kobiljonov Davlatbek Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon
6.	14th Week 9th Dec. to 15th Dec. 2024	All Teams: Integration of all components, System testing, Bug fixing, Performance optimization Specific Tasks: Frontend-Backend integration, Socket communication testing, Database performance testing All Teams: Final system testing, User acceptance testing, Performance optimization, Documentation updates, Bug fixes and refinements	Frontend: Sabitov Samandar, Kobiljonov Davlatbek Middle Layer: Xatamov Javohirbek, Eshboyev Temur Backend: Zikirillayev Asadtilla, Sodikov Akmaljon
7.	15th Week 16th Dec. to 22nd Dec. 2024	FINAL EXAM WEEK Term Project Final Report Preparation	
8.	16th Week 23rd Dec. to 28th Dec. 2024	- Term Project Final Report Preparation and Submission - Term Project Presentation	All members participating in presentation and demo. Sabitov Samandar, Kobiljonov Davlatbek, Xatamov Javohirbek, Eshboyev Temur, Zikirillayev Asadtilla, Sodikov Akmaljon.

PROJECT TEAM :

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Team Leader :

Name	Student ID	Signature with date
1. Jaschirbek Xatamov	U2210251	<i>[Signature]</i> 24.10.2024

Team Members :

Name	Student ID	Signature with date
2. Zikizillaev Asadilla	U2210280	<i>[Signature]</i> 24.10.2024
3. Sodirov Akmorlon	U2210214	<i>[Signature]</i> 24.10.2024
4. Eshboyev Temur	U2210068	<i>[Signature]</i> 24.10.2024
5. Sabitov Samondar	U2210197	<i>[Signature]</i> 24.10.2024
6. Kobiljonov Davlatbek	U2210718	<i>[Signature]</i> 24.10.2024

Monday, October 21, 2024

REFERENCES:

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- [2] The GTK+ Team, "GTK+ 3 Reference Manual," GNOME Developer, 2024. [Online]. Available: <https://docs.gtk.org/gtk3/>.
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- [4] R. Love, "Linux System Programming: Talking Directly to the Kernel and C Library," 2nd ed. Sebastopol, CA: O'Reilly Media, 2013.
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- [6] D. P. Bovet and M. Cesati, "Understanding the Linux Kernel," 3rd ed. Sebastopol, CA: O'Reilly Media, 2005.